

Airline Booking System

Discussion

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The Significant Problem:

Deciding on the Brief:

For my coursework, I have elected to build a booking system for a fictional airline (Infinity Airways).

Going into my coursework, I was tempted to take the easy route. I thought that it would be best for me to take the route to simple and guaranteed marks with a to-do list tracker that I am confident would be within my gift. The only way that this would have pushed me in any way would have been in the automatic printing of the day's itinerary at the beginning of each day. Having consulted with my teacher, we agreed that this was beneath me.

Moving on from this, I had to put a great deal of thought into finding a project that was sufficiently advanced and captivating while remaining within what I could feasibly do with the time I have. I have had a certain affinity for the aviation industry for almost five years now and was certain that this would be the direction that I would want to take things in. From this, I started to think that I could build a system to track the schedules and locations of the different aircraft and staff of an airline. This would have been interesting but largely without depth. This would come as I would have no way of interacting with the locations of real world aircraft (I could track them but could not assign them to set routes or times). This would mean that everything would essentially be on rails with each aircraft being exactly where you expect it to be at each given point in time (no delays or unexpected maintenance requirements). If I wanted to include this level of depth, it would have to be implemented such that events would happen randomly. This had an unsavoury taste of gamifying my software that I wasn't particularly fond of.

Hoping to stick with the aviation theme, I then moved on to thinking about making the booking system for an airline. This is still a large project with my having to implement a detailed relational database structure for bookings and how to schedule repeating flights efficiently. It would also give sufficient opportunity for additional developments with surge pricing and advanced route searches. I hope that this project will push me in a way that not only demonstrates the extent of my ability for a fair grade that represents me but also to push and develop that ability.

With this idea, I was initially planning to build a booking system for British Airways (my dream airline to work for). I was then forced to reconsider this on the fact that British Airways already has a website for this exact function. This would force me to be constrained by their design methodologies and take excessive inspiration from them. In effect, all of my designs would be done for me with all the work left for me to do being screenshotting the existing website. This is not the direction that I want to take my project in, so I have elected to make my website for a fictional airline that I used to fly for in my flight simulator: Infinity Airways. This is a great middle ground where I don't have to come up with a completely new brand identity and I am presented with an existing colour palette but I still have free reign with the direction I want to take the project in. I believe that this is reminiscent of the situation I would be presented with should I go into software development as a career; clients may already have an established brand identity but I will still have to do the heavy lifting in the design department if I am making an entirely new system. Another benefit of this is that there are a large number of free liveries for my flight simulator that I will be able to use to take screenshots for the website.

Meet The Client:

Infinity Airways is a brand-new airline that has received a large investment to start up. It is expecting to scale quickly and run a large number of flights from its launch. It is a luxury airline targeting the segment of the market in between airlines and private aircraft ownership / charter.



Infinity Airways differentiates itself through their incredible attention to detail and sense of luxury at a price that rivals other high-end airlines. In this vein, each customer is referred to as a 'guest' and has an assigned personal support representative.

Despite the incredible luxury of the airline, it does not make this a main selling point. This serves more as a given with focus instead being placed on the destination. Infinity Airways sees itself more as a facilitator of incredible experiences as it respects that its clientele will likely have limited spare time available for holidays. Most branding for the airline is focused on the streamlined nature of the airline and the trust that you can place in it.

While the airline focuses its advertising on holiday destinations, it also serves a number of business hubs in North America, The UAE, Asia, and Europe. While these are not advertised as extensively as other routes, Infinity Airways expects flights to be in high demand as guests of holiday flights are impressed by the reliability and understated luxury of the airline and elect to use them for their business trips too.

With its limited advertising focusing on luxury, Infinity Airways runs the risk of blending in with other airlines that serve the same destinations for a lower price. As such, the airline expects to be more for "those who know" in high paying and executive business roles. As such, it will draw elements from private golf club memberships and other exclusive entities. This will likely be given attention in the branding and marketing of the airline.

Finally, the airline has taken note of growing distaste towards the over commercialisation of air travel with extortionate credit cards and the collection of meaningless points. As such, guests do not collect "miles" or any equivalent. While this may seem strange in the current climate that focuses on savings with ever growing costs of living, this is presented as a positive in the branding of the airline.

Target Audience Profiles:

Person #1

Income: £150,000

Age: 58

Profession: Banking

This individual has a stressful schedule that can be hard to manage at times. They get a limited number of days off each year and is taking an extended weekend away in a tropical location for a holiday. They will likely work on their trip. They will likely buy an economy seat.

Person #2

Age: 28

Profession: Early Retired

This individual made their fortune having sold their startup and is looking for a fun time full of excitement/adventure. Frequently charts for larger groups but is travelling on own on this occasion. They will likely buy a first-class seat.

Person #3

Income: £850,000

Age: 62

Profession: Business Executive

This individual is on a business trip from London to New York for a convention. The trip is being treated as a business expense and they will likely buy a business class seat. They will most likely work on the plane and will prioritise minimal disruptions. They would favour a night flight so that they do not lose any daylight hours they could be working in.

The Client Brief:

Types of users:

- Guest (logged out),
- Guest (logged in),
- Flight operations manager,
- Check-in agent,
- Gate agent,
- Customer support representative,
- Admin,

Logged out guests will have limited capabilities. The main function of this side of the website is to funnel prospective guests to booking a flight/holiday (which will require their being logged in). The page here will be virtually identical to that of the logged in user except that they will be prompted to create an account / log in as they try to finalize their booking. A guest here will want the booking information they have already filled out here to persist so that they do not have to re-enter it. This will make the transaction as frictionless as possible leading to increased sales and a closer adherence to the core values of Infinity Airways. The key needs of logged out guests will be:

- Create account / log in,
- Search for flights and holidays,
- Enter preliminary information to book flights,
- Booking information persists after log in,

Logged in guests will have similar needs to logged out ones except they will be able to view their personal dashboard, book flights, and edit personal information. These users' primary needs will be:

- Search for flights and holidays,
- Book flights,
- Edit personal information (billing information, email, name, password),
- Cancel booked flights,
- Edit booked flights,

Flight operations managers will be the users adding routes, services, and flights as well as editing them. Their primary needs will be:

- Create routes,
- Add services to routes,
- Add one-off flights between destinations / for routes,
- Edit routes, schedule, and flight information,
 - Delay,
 - Cancel,
 - Update details,
- View an overview of information about flight statistics,
 - They should not be able to see individual passengers booked on to flights but instead an abstracted version detailing passenger numbers on individual flights / routes compared to capacity. They should also be able to see an overview of passengers over time (graph). This will allow them to assign resources where they are most needed without invading the privacy of guests;

- Add articles for holiday destinations that direct to flights,

Check-in and gate agents are two peas of a pod. They will both have to see an overview of all flights for a given day at a specific airport. They will then have to see the passengers assigned to each flight and be able to change their state between preset options (e.g.: pending arrival, checked in, boarded, deboarded). Check in agents may have additional privileges to edit booking information such as seats (including upgrading class) – this can be charged or for free. Their primary needs will be:

- View active flights at a given airport,
- View passengers on active flights at the given airport,
- Change state of passengers for active flights at given airport,
- (In the case of privileged check-in agents) change booking information such as class and seats,
 - Give option to charge for upgrade or give for free,

Customer support representatives will act as the spokespeople of the airline as they help guests with their bookings. Each guest will be assigned a personal support rep (in keeping with the luxury theme of the airline). They will have complete read and write authority over the bookings of the guests assigned to them. This will include booking new flights for them as well as giving information about or altering existing ones. As a member of staff, the UI for these users can be more crowded as they will most likely receive specialist training in the software. The primary needs of these users will be:

- View all guests assigned to them,
- View the bookings of guests assigned to them,
- Edit the bookings of guests assigned to them,
- Create bookings for guests assigned to them,

The airline will also have a number of admin users. These will be the users that add new users and change their staff type. They will also be able to delete the accounts of users if needed. The primary needs of these users will be:

- View all current staff and guest accounts,
- Delete accounts,
- Change staff account types,
- Create staff accounts,
- Assign additional privileges to staff (e.g.: give check-in agents rights to upgrade users),

Broad Aims & Objectives:

The above lays out an overview of what the program will ideally do. I find it more helpful to split this into three distinct categories of priority: threshold, target, and stretch. Threshold (MVP) contains the most critical functions for the tool. I then build on this with target which contains the elements that I would like and expect the program to do by the deadline. Finally, I top this all off with stretch. This contains all of the things that I would love to be able to implement while realising that they may not be feasible within the time limit while maintaining a high standard of code quality. As such, they (ideally) will only be implemented when all preceding aims and objectives are complete to an exemplary standard. Within the headers, bullet points are in no particular order.

Threshold (MVP):

1. Flight operations users can add flights,
2. Guest users can book on to flights,
3. Brought together with a GUI,
4. Make use of a database containing 4 tables in third normal form,

Target:

1. Use secure login using user sessions (Flask),
2. Flight scheduling functionality,
3. Search for flights,
4. Guest user accounts,
 - a. Guests can view their previously booked flights,
5. Customer support representative accounts,
6. Check-in & gate agent accounts,
7. Flight operations managers can see performance of flights / routes / services over time,
 - a. As a graph,
8. Guests can specify the seat they want to book within the GUI,
9. Guests can specify the identities of other passengers in their group,
 - a. Passport information,
10. Admin users can add new staff users,
11. Admin users can view and search guest accounts,
12. Admin users can delete guest accounts,
13. Admin users can give staff accounts special privileges,
14. PDF boarding pass is generated,
 - a. Can be printed / downloaded by guest, customer support representative, and check-in agent,
15. Flight operations users can cancel, edit, and delete flights / routes / services,
16. Guests can book lounge access,
17. System will automatically generate a callsign for each flight (unique for its time slot),

Stretch:

1. Email verification,
2. Email boarding passes,

3. Email password reset for guest accounts,
4. Use Stripe test mode for mock financial transactions,
5. Flight operations users can add articles for destinations,
 - a. Direct to flights,
6. Advanced search for flights,
 - a. Displays “similar flights” that depart from / arrive at nearby airports,
 - i. Use a quick sort algorithm to order by how close they are to entered airports,
7. Add surge / dynamic pricing,
8. Autofill details during the booking process for logged in users,
 - a. Credit card details,
 - b. Previous companions,
9. Staff accounts can interact as guests with special discounts / offers,
 - a. They can book on to empty flights at very short notice,
10. Different aircraft configurations have different seat maps,
11. Flight operations managers can add new airports that are not in the database,
12. Flight operations managers can delete and update existing airports,

When I start my development, I will start with more fundamental elements while keeping mind of the future developments that will come. This will ensure that my program is easily maintainable and scalable as I add on elements from the target and stretch categories. That being said, I will not start by just making the MVP. I feel that this would be unnecessarily limiting and slow the development of the system more than is reasonable. I will most likely start by making a basic outline that combines all of the threshold elements with a large number of the target elements that I feel would be simple enough to implement and / or harder to implement after the fact if it would require large alterations to the codebase when it could realistically just be done in the first pass.

Limitations of the Solution Presented:

While I believe that the solution I have presented would be helpful for the hypothetical client, it goes without saying that it will not be without limitations and flaws. For example, I will likely not be able to implement a fleet management system. This will mean that the system will have to always assume that there are sufficient aircraft at each airport for the flights that originate from it. Flights will also not be automatically scheduled such that only one flight departs per flight that arrives. This will be the responsibility of the flight operations managers to coordinate. This could mean that routes, services, and flights are made when there would be no aircraft at the location to service it.

The implementation of a fleet management system would be a great addition to this tool but likely falls outside of the scope of what is possible with the time allotted; this would be the workload of an entire extra project tacked on top of what is already likely to be quite a demanding project.

Similarly, I will not be managing staff rotas or assignment with this system. The system will assume that all aircraft will always be fully crewed and that all crew arrive on time (flight operations managers can manually delay flights if needed).

For clear reasons, I will also not be handling any money with my system. This goes beyond what is needed for coursework and goes beyond my level of comfort. One of my stretch targets is to implement Stripe test mode for mock transactions. This will provide me with fake card details that will simulate successful, insufficiently funded, and declined transactions. This will put me as close to realism as is reasonable and (if this were ever to be converted to a program for a real client) allow for a quick and easy changeover by simply changing the API key.

Furthermore, this booking system will have to run as a website. This means that it will have to run on a server 24/7. This may place strain on the server at expense to the client if a large number of users are using the website at the same time. Furthermore, when there are many users using the website at the same time, the website may grow slower as the server struggles to keep up. This may reduce the quality of the experience for guests leading to a discrepancy between the brand aims and the experience of the user. This may also happen when scheduled tasks take place such as the creation of flight entries from schedules at a set time each day. When there are large quantities of services, this will likely place a lot of strain on the servers. In a real-world scenario, this may be placed on a separate server to prevent interruptions to guests. Kubernetes and a load balancer could also be used to spread load between a collection of high-performance servers helping to ensure that all elements of the system have the resources they require but this would only be reasonable at a large scale that this project will not reach during testing.

The system will need to comply with the data protection legislation of the territories that it operates in. I will take measures to comply with many of these mandates, but a full independent review of data protection compliance would be required if this were a real product. Under GDPR, I would be able to keep details of bookings and flights for as long as a user maintains an account. This would fall under “legitimate interest” as previous booking information can assist in the record keeping for users as well as allow them to rebook holidays. If a user deletes their account, I can delete their user account and replace any foreign keys that reference it with null (the process of nullification) to comply with the right to erasure under UK GDPR. This would automatically anonymise records without deleting them as is allowed within the constraints of UK GDPR.

Details of payment amount, date, and the VAT code of bookings can be retained in compliance with HMRC law which mandates payment records must be kept for at least 7 years even though the details of who paid don't need to be retained. The CAA also mandates that airlines must retain operational records for a set period of time. This includes the passport numbers of passengers on the manifest for safety and security purposes. This will mean that I will not have to delete passport data when a user account is deleted as long as they have flown within that period. One of the areas where my system will likely not be fully compliant will be the deletion of the passport data after the CAA mandated period or if they did not fly. This would have to be fixed if the system were ever adapted for a real client. As mentioned previously, a full review would be required to ascertain the territories that interact with the system and how the system complies with their respective data protection laws.

The system I have proposed also does not include a mobile app. While I will endeavour to make my web UI responsive to different screen sizes, I will not be making a standalone app. A standalone app could be helpful for the client as it would allow them to send notifications to prompt purchases (Pegasus Airlines saw that users who received push notifications purchased 72% more tickets – [Source: dataroid](#)). This may be something to be considered if I were to further develop this project.

The system will also not report on the profitability of flights as the costs will be unknown to it due to the ever-changing price of fuel and other operating expenses. Instead, it will only report the revenue of each flight and flight operations managers, accountants, and executives will have to independently decide if they are worth continuing.

Feedback

#1 – Can you group articles?

Articles were one of the items I mentioned as a stretch targets. They will likely select a collection of almost favoured articles that will show on a carousel on the home page. Other than these, there will be a page that guests can go to in order to view all articles which will likely be grouped by destination type (e.g.: tropical, cold, ocean, etc.).

#2 – What will you show in the admin page?

The main admin dashboard will just show all of the different users and their associated privileges. Admin users will then be able to edit user details and privileges from this dashboard. Admin users will also be able to enter all the other portals with their login credentials and will have full privileges but this will not be implemented into their main dashboard beyond perhaps some redirect links.

#3 – Will you have a frequent flier program?

I had initially thought against this as it may build to a sense of over commercialisation of the airline industry but was told that it is possible that the richer clientele of Infinity Airways may instead prefer this as it allows them to 'hold on to their money.' This will also give me more opportunities to build complexity into my program so I will likely be adding this to the target list of aims and objectives.

#4 – How will you set user privileges?

This links closely into feedback point #2. Admin users will be able to tick relevant privileges for users which will change Boolean fields for each privilege in the users table.

#5 – Are passengers just booking seats?

While the main function of the program is to facilitate the booking of seats, it is a target to also include the booking of lounges. I may also be able to add the booking of meals and baggage along with other similar extras.

#6 – If they have to change airports, will you offer transport?

Guests are free to contact their customer support representative if they have to change flights but this is not something that this bit of software has to manage.

#7 – Will you show cheaper / better flights?

The richer clientele with limited time will likely prioritise faster flights than cheaper ones but I could add it as a stretch target to make a shortest path algorithm to find the cheapest route to a location with changes.

#8 – Will you show different times?

Guests will be able to enter their search criteria and any flights that match it will be returned. If no date / time window is provided, all flights at all times for the route will be given.

#9 – Will you show stops / direct flights?

With the stretch target mentioned in feedback point #7, users will be able to see the number of changes from the search and click on the flight for more details about where changes are.

#10 – Will you show all flights?

I feel that showing all flights could easily be overwhelming but showing all routes could be helpful for those that don't know where they want to go on holiday. These could be grouped by category similar to the articles in feedback point #1.

Amended Aims & Objectives:

New additions to the aims and objectives are marked in **bold**.

Threshold (MVP):

5. Flight operations users can add flights,
6. Guest users can book on to flights,
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8. Make use of a database containing 4 tables in third normal form,

Target:

18. Use secure login using user sessions (Flask),
19. Flight scheduling functionality,
20. Search for flights,
21. Guest user accounts,
 - a. Guests can view their previously booked flights,
22. Customer support representative accounts,
23. Check-in & gate agent accounts,
24. Flight operations managers can see performance of flights / routes / services over time,
 - a. As a graph,
25. Guests can specify the seat they want to book within the GUI,
26. Guests can specify the identities of other passengers in their group,
 - a. Passport information,
27. Admin users can add new staff users,
28. Admin users can view and search guest accounts,
29. Admin users can delete guest accounts,
30. Admin users can give staff accounts special privileges,
31. PDF boarding pass is generated,
 - a. Can be printed / downloaded by guest, customer support representative, and check-in agent,
32. Flight operations users can cancel, edit, and delete flights / routes / services,
33. Guests can book lounge access,
34. System will automatically generate a callsign for each flight (unique for its time slot),
35. **Guests can collect miles for airline bonuses,**
36. **Guests can book lounges,**
37. **Guests can book baggage,**
38. **Guests can book meals,**

Stretch:

13. Email verification,
14. Email boarding passes,
15. Email password reset for guest accounts,
16. Use Stripe test mode for mock financial transactions,
17. Flight operations users can add articles for destinations,

- a. Direct to flights,
 - b. **Grouped by types of destination (e.g.: tropical),**
18. Advanced search for flights,
- a. Displays “similar flights” that depart from / arrive at nearby airports,
 - i. Use a quick sort algorithm to order by how close they are to entered airports,
19. Add surge / dynamic pricing,
20. Autofill details during the booking process for logged in users,
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 - b. Previous companions,
21. Staff accounts can interact as guests with special discounts / offers,
- a. They can book on to empty flights at very short notice,
22. Different aircraft configurations have different seat maps,
23. Flight operations managers can add new airports that are not in the database,
24. Flight operations managers can delete and update existing airports,
25. **Guests can give allergy information,**
26. **Add a shortest path algorithm for cheapest route when searching,**